Done

Musical Tracks Many-to-One

This application will read an iTunes library in comma-separated-values (CSV) format and produce properly normalized tables as specified below. Once you have placed the proper data in the tables, press the button below to check your answer.

Host: pg.pg4e.com

Port: 5432

Database: pg4e_3774c65c52

User: pg4e_3774c65c52

Password: (hide/show copy)

Check Answer

Here is the the **psql** command to access your database:

```
psql -h pg.pg4e.com -p 5432 -U pg4e_3774c65c52 pg4e_3774c65c52
```

Here is the structure of the tables you will need for this assignment:

```
CREATE TABLE album (
  id SERIAL,
  title VARCHAR(128) UNIQUE,
  PRIMARY KEY(id)
);

CREATE TABLE track (
    id SERIAL,
    title VARCHAR(128),
    len INTEGER, rating INTEGER, count INTEGER,
    album_id INTEGER REFERENCES album(id) ON DELETE CASCADE,
    UNIQUE(title, album_id),
    PRIMARY KEY(id)
);

DROP TABLE IF EXISTS track_raw;
CREATE TABLE track_raw
 (title TEXT, artist TEXT, album TEXT, album_id INTEGER,
  count INTEGER, rating INTEGER, len INTEGER);
```

We will ignore the artist field for this assignment and focus on the many-to-one relationship between tracks and albums.

If you run the program multiple times in testing or with different files, make sure to empty out the data before each run.

Done

Load this CSV data (library.csv?PHPSESSID=4e5dd2822b534a5153210a093f4c2865") file into the **track_raw** table using the **\copy** command. Then write SQL commands to insert all of the distinct albums into the **album** table (creating their primary keys) and then set the **album_id** in the **track_raw** table using an SQL query like:

```
UPDATE track_raw SET album_id = (SELECT album.id FROM album WHERE album.title = track_raw.album);
```

Then use a **INSERT ... SELECT** statement to copy the corresponding data from the **track_raw** table to the **track** table, effectively dropping the **artist** and **album** text fields.

To grade this assignment, the auto-grader will run a query like this on your database and look for the data it expects to see:

```
SELECT track.title, album.title
    FROM track
    JOIN album ON track.album_id = album.id
    ORDER BY track.title LIMIT 3;
```

The expected result of this query on your database is:

track	album
A Boy Named Sue (live)	The Legend Of Johnny Cash
A Brief History of Packets	Computing Conversations
Aguas De Marco	Natural Wonders Music Sampler 1999